MILITARY SPECIFICATION
INVERTERS, AIRCRAFT, GENERAL SPECIFICATION FOR

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope - This specification covers devices, for use on aircraft, which convert power from 28-volt dc systems to ac power at 400 cps.

1.2 Classification - Inverters shall be of the following classes, as specified:

Class A - Inverters which are self-cooled and designed to meet the altitude-temperature requirements of Standard MS33543, curve II.

Class B - Inverters which are self-cooled and designed to meet the altitude-temperature requirements of Standard MS33543, curve I, except that the ambient temperature shall not exceed 85°C. The inverter shall deliver rated output up to 50,000 feet and 50 percent (minimum) rated power to 65,000 feet.

Class C - Inverters which are blast-cooled and designed to meet the altitude-temperature requirements of MS33542.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416 Plating, Cadmium (Electrodeposited)

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Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy; Processes for Corrosion Protection of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-100	Engineering drawing practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS26515	Plate, Identification, Use Data
MS33542	Criteria - Temperature and Altitude Range Blast Cooled Electric Equipment
MS33543	Criteria - Temperature and Altitude Range Self Cooled Electric Equipment
MS33586	Metals, Definition of Dissimilar

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Federal

FED-STD-151 Metals, Test methods

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - The inverters furnished under this specification shall be a product which has been tested, and has passed the Qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Materials - Materials used in the manufacture of units shall be suitable for the purpose and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used provided the specifications are approved by the Government and contain provisions for adequate tests. The use of the contractor's specifications will not constitute waiver of Government inspection.

3.2.1 Metals - Metals shall be of a corrosion resistant type or suitably processed to resist corrosion.

3.2.1.1 Dissimilar metals - Unless otherwise suitably protected, dissimilar metals, such as brass, copper, or steel, shall not be used in intimate contact with magnesium, aluminum or their alloys. When protection is used, it shall be of such type that a low-impedance path is offered to radio-frequency currents. Dissimilar metals are defined in Standard MS33586.

3.2.2 Non-metallic materials - Non-metallic materials used shall be flame resistant and shall not support combustion.

3.3 Selection of materials - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification shall be selected in accordance with Standard MIL-STD-143, except as provided in the following paragraph.

3.3.1 Standard Parts - Military standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Any standard part, of the same part number, shall be readily inter-

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changeable as applied within the unit. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc. may be used, provided they possess suitable properties and are replaceable by the Military standard parts without alteration, and provided the corresponding Military standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable Military standard part in effect on date of invitation for bids, equivalent commercial parts other than electron tubes and semiconductor devices may be used provided they conform to all requirements of this specification. For electron tubes and semiconductor devices, in the event there is no suitable Military standard part in effect on date of invitation for bids, permission for a substitute part shall be obtained from the procuring activity.

3.4. Design and Construction -

3.4.1 Detail specifications and drawings - Units shall conform to the requirements on the applicable Air Force - Navy - Aeronautical Standard (AN) or Military Standard (MS) drawing or detail specification. Whenever the requirements of this specification and the detail specification or drawing conflict, the requirements of the detail specification or drawing shall govern. Any deviation from this specification or from subsidiary specifications or drawings, where applicable, must be specifically approved by the procuring activity.

3.4.2 Simplification - Simplicity of design resulting from use of the same part for as many applications as possible in a given unit and in units of different ratings, is highly desirable.

3.4.3 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-STD-100.

3.4.4 Electrical connections - Internal wiring of the unit shall be in accordance with the applicable drawing or detail specification and shall terminate as shown thereon. If input and output contactors are used, they shall be de-energized simultaneously.

3.4.5 Screw threads - Screw threads shall conform to the requirements of MIL-S-7742.

3.4.6 Safety wiring and staking - Accidental loosening of screws and screw parts and other connections shall be prevented by safety wiring (0.032-inch minimum OD) where practicable, staking or other approved method. Washers and

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cotter pins, where used, shall be assembled in a manner which prevents rotation of washers and movement of cotter pins under conditions of vibration.

3.4.7 Rotor - Rotating type units shall be so constructed that the rotor may be removed readily without special tools.

3.4.8 Brushes (Brush type units only)-

3.4.8.1 Brush accessibility - Brushes shall be accessible for replacement when brush opening covers are removed. Brush opening covers shall be easily accessible and removable without special tools.

3.4.8.2 Brush fit - The face of each commutator brush shall contact its commutator 100 percent in the direction of rotation for at least 75 percent of brush dimension parallel to the shaft. The face of each slip ring brush shall contact its slip ring over an area that will insure satisfactory performance of the inverter; however, as a minimum requirement, slip ring brush contact area shall be 50 percent.

3.4.8.3 Cartridge-type brushes - When cartridge-type brush assemblies are used, there shall be an integral assembly of the brush, spring, and terminal which shall not rotate during installation or during service.

3.4.8.4 Brush position - Brush riggings for all units of the same part number made by one manufacturer, shall be completely interchangeable. Brush rigging in each inverter shall be pinned in that position which results in optimum commutation characteristics for that unit. The method of pinning shall not adversely affect interchangeability of parts between units, and shall allow the brush rigging position to be reset during overhaul.

3.4.8.5 Brush marking - Each brush shall be marked for allowable wear. Brushes having a cross sectional area parallel to the wearing surface larger than 1/16 square inch shall be marked for allowable wear by a readily discernible groove in its edge. Brushes having cross sectional area parallel to the wearing surface of less than 1/16 square inch may be marked with a line. In either case the line or groove shall be as shown in Fig. 1.



BRUSH MARKINGS
FIGURE 1.

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3.4.8.6 Spare brushes - All spare brushes shall be formed to a radius equal to approximately 100 per cent of the radius of the new commutator or slip ring and shall be of the same specifications as the corresponding brushes approved by the Qualification tests.

3.4.9 Protection - All ventilating openings shall be of such size as to prevent passage of 5/16-inch diameter rod.

3.4.10 Vibration isolators - Vibration isolators (if used) shall be of such design that the main support between the stationary and isolated member is of all metal construction.

3.4.11 Input voltage - The nominal input voltage shall be 28 VDC, or as specified on the applicable MS drawing.

3.4.12 Adjustment - No adjustment other than for voltage and frequency (as provided for by 3.4.12.1 and 3.4.12.2) shall be required to put the unit into satisfactory operation at the time of installation on the aircraft.

3.4.12.1 Voltage setting - When required by the applicable MS drawing, the unit shall be constructed to permit the output voltage to be set over a range of at least ±5.0 percent of rated value in increments of not more than 0.5 percent of rated voltage. The setting means shall be so constructed that it will not be displaced by vibration or shock. The control shall be designed for adjustment using a straight screwdriver.

3.4.12.2 Frequency setting - No frequency adjustment will be required if the control device is of such construction as not to require adjustment during the life of the equipment. If frequency adjustment is incorporated in the unit, the control shall be so constructed that it will not be displaced by vibration or shock, and that it shall be adjustable with a straight screwdriver.

(a) Units 250 VA and larger shall have an adjustable range at least ±2.5 percent of the rated frequency in increments of not more than 0.25 percent of the rated frequency.

(b) Units smaller than 250 VA shall have an adjustment range of at least ±5.0 percent of the rated frequency in increments of not more than 0.25 of the rated frequency.

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3.5 Performance -

3.5.1 Operating conditions - The unit shall deliver power in accordance with the applicable MS drawing and this specification.

3.5.2 Overload - The unit shall be capable of delivering 200 percent rated load current for 5 seconds at the minimum voltage output shown on the applicable drawing. The unit shall be capable of delivering a short circuit current of at least 250 percent rated load current.

3.5.3 Lubrication - The unit shall not require additional lubrication during 1000 hours of operation.

3.5.4 Brush life - Brushes when used shall have a useful life of no less than 500 hours.

3.5.5 Life - The unit, when operating under any temperature or altitude conditions indicated by the detail specification or drawing, shall have a useful life of no less than 1000 hours. The brushes (when used) may be replaced after 500 hours of operation.

3.5.6 Efficiency - The efficiency of the unit shall be within the limits specified on the applicable MS drawing.

3.5.7 Starting time - At a time no more than 2 seconds after the input terminals are energized with specified input voltage, the output voltage and frequency shall be within and remain within the limits specified on the applicable MS drawing. If time is required for the voltage and frequency controls to become effective, the uncontrolled voltage shall not exceed the limits shown in Figure 2, and the frequency shall not exceed 105 per cent of the maximum controlled value specified by the applicable MS drawing after 0.25 second.

3.5.8 Starting contactor - A starting contactor, is used, shall not open at a voltage higher than 9 volts DC at any temperature between -55°C and the maximum temperature indicated by the applicable MS drawing.

3.5.9 Starting current - The starting current shall not exceed 1000 percent of full load current as computed from the minimum allowable efficiency specified on the applicable MS drawing or 1500 amperes, whichever is less. The current shall return to and stay within 300% of full load value within 0.3 second after first application of input voltage.

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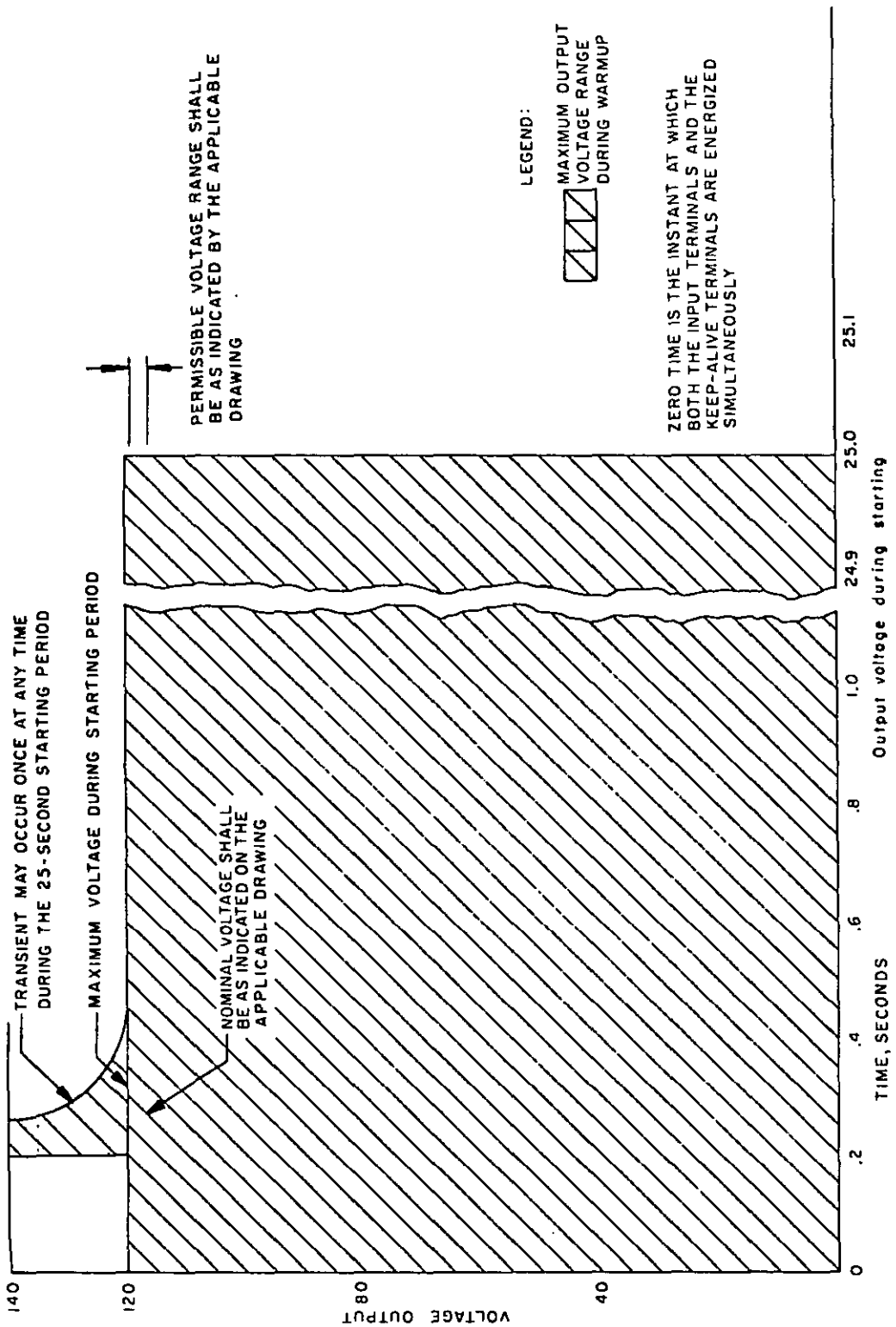


FIGURE 2.

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3.5.10 Radio interference - Enclosing case construction shall provide continuity of electrical shielding with a low radio-frequency impedance path to ground and across all mechanical discontinuities. Conducted and radiated interference produced by the operation of the unit shall not exceed the requirements of Specification MIL-I-6181.

3.5.11 Environmental conditions - The unit shall withstand the following environmental conditions with no degradation of performance:

(a) Sand and dust

(b) Salt laden atmosphere

(c) Humidity

(d) Fungus

(e) Shock

(f) Vibration

(g) Acceleration

(h) Unit shall conform to this specification when operated in any position.

3.5.12 Phase rotation - Unless otherwise specified, phase rotation shall be A, B, C.

3.6 Finish -

3.6.1 Aluminum or aluminum-alloy parts - Where practicable, unless of corrosion resistant types aluminum-alloy parts shall be treated for corrosion resistance in accordance with Specification MIL-A-8625 or Specification MIL-C-5541. The film deposited by this treatment shall be removed from the actual contact area of all surfaces required to act as a path for electrical current and from the local areas under screws, nuts, or the like used for assembly or mounting purposes, to provide an adequate bonding connection.

3.6.2 Magnesium-alloy parts - Wherever practicable, magnesium-alloy parts shall be surface treated in accordance with Specification MIL-M-3171 to provide protection against corrosion.

3.6.3 Plating - Cadmium plating shall be in accordance with Specification QQ-P-416.

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3.7 Markings -

3.7.1 Terminals - Markings of terminals shall be in accordance with the applicable drawing or detail specification.

3.7.2 Rotation marking - On rotating-type units, the direction of rotation shall be plainly indicated by an arrow.

3.7.3 Wiring diagram - Attached to the inside of the control box cover or other surfaces, when practicable, there shall be a circuit diagram with a legend describing the electrical components of the equipment. This diagram shall include rated values of resistors and capacitors. The diagram shall be legible, protected against humidity, and not easily removed.

3.7.4 Maintenance information - A use data plate in accordance with Standard MS26515 shall be attached to the inverter.

3.7.5 Air flow marking - When applicable, the direction of air flow shall be indicated by an arrow.

3.7.6 Identification of product - Equipment, assemblies, and parts shall be marked in accordance with Standard MIL-STD-130. The identification data applied to the inverter shall be as follows:

Nomenclature (as shown on applicable drawing)

Input:_____Volts DC_____Amperes DC

Output:_____Volts AC_____Amperes AC
_____CPS_____Phase

MS Part No.
Federal Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's name or trade-mark
Inspector's stamp
US

If the MS drawing provides for the speparation of the unit into a control component and a power component, each component shall be equipped with a nameplate bearing the above information plus the following:

To be used with (control or power component)*

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Stock No.
Manufacturer's part No.

*Manufacturer to insert applicable nomenclature.

3.8 Workmanship - All machined surfaces shall have a smooth finish, and all details of manufacture, including the preparation of parts and accessories, shall be in accordance with the best practices for high-quality electrical equipment. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts, plating, lacquering, riveting, clearance between soldered connections, and ruggedness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of tests - The inspection and testing of inverters shall be classified as follows:

(a) Qualification tests: Qualification tests are those tests performed on samples submitted for approval as qualified products.

(b) Acceptance tests: Acceptance tests are those tests performed on individual lots which have been submitted for acceptance.

4.3 Qualification tests -

4.3.1 Sampling instructions - The Qualification test samples shall consist of six units of each part number upon which qualification is desired. The manufacturer shall furnish all samples and shall be responsible for accomplishing the required tests on four samples.

4.3.2 Tests - The Qualification tests shall consist of all the tests of this specification, as described under "Test methods." The tests shall be conducted on each of four units as shown below and must be conducted essentially in the order listed for each unit.

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UNIT NO. 1

Tests	Load Condition
(a) Examination of product	Load not applicable
(b) Heating, overload, and commutation (commutation applicable to rotating-type units only).	1 Ø
(c) Dielectric strength	Load not applicable
(d) Voltage and frequency regulation	3 Ø Wye
(e) Efficiency	3 Ø Wye
(f) Phase balance (3 Ø units)	3 Ø Wye
(g) Output modulation	3 Ø Wye
(h) Wave form	3 Ø Wye and 1 Ø
(i) Reduced input voltage	3 Ø Wye
(j) Overspeed (applicable to rotating type units)	3 Ø Wye
(k) Radio interference	3 Ø Wye and 1 Ø
(l) Factors affecting voltage and frequency regulation	3 Ø Wye
(m) Acceleration	3 Ø Wye
(n) Recovery time	3 Ø Wye
(o) Starting	3 Ø Wye
(p) Operating position	3 Ø Wye
(q) Explosion-proof (when specified on applicable drawing)	1 Ø
(r) Endurance	1 Ø
(s) Shock	3 Ø Wye

UNIT NO. 2

(a) Examination of product	Load not applicable
(b) Heating, overload, and commutation (commutation applicable to rotating type units only).	1 Ø
(c) Dielectric strength	Load not applicable
(d) Voltage and frequency regulation	1 Ø
(e) Efficiency	1 Ø
(f) Phase balance (3 Ø units only)	3 Ø Wye
(g) Output modulation	1 Ø
(h) Reduced input voltage	1 Ø
(i) Wave form	3 Ø Wye and 1 Ø
(j) Overspeed (applicable to rotating type units only)	1 Ø
(k) Radio interference	3 Ø Wye and 1 Ø
(l) Factors affecting voltage and frequency regulation	1 Ø

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Tests	Load Condition
(m) Acceleration	1 Ø
(n) Recovery time	1 Ø
(o) Starting (oscillograms only)	1 Ø
(p) Operating position	1 Ø
(q) Performance	1 Ø
(r) Humidity	1 Ø
UNIT NO. 3	
(a) Examination of product	Load not applicable
(b) Heating, overload, and commutation (commutation applicable to rotating type units only).	1 Ø
(c) Dielectric strength	Load not applicable
(d) Voltage and frequency regulation (room temperature only)	3 Ø Wye
(e) Efficiency	3 Ø Wye
(f) Output modulation	3 Ø Wye
(g) Wave form	3 Ø Wye
(h) Phase balance (3 Ø units only)	3 Ø Wye
(i) Overspeed (applicable to rotating type units only)	3 Ø Wye
(j) Performance	3 Ø Wye
(k) Vibration	3 Ø Wye
(l) Fungus	3 Ø Wye
(m) Salt spray	3 Ø Wye
UNIT NO. 4	
(a) Examination of product	Load not applicable
(b) Heating, overload, and commutation (commutation applicable to rotating type units only).	1 Ø
(c) Dielectric strength	Load not applicable
(d) Voltage and frequency regulation (room temperature only)	1 Ø
(e) Efficiency	1 Ø
(f) Output modulation	1 Ø
(g) Wave form	3 Ø Wye and 1 Ø
(h) Phase balance (3 Ø units only)	3 Ø Wye
(i) Overspeed (applicable to rotating type units only)	1 Ø
(j) Life	1 Ø
(k) Sand and dust	1 Ø

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4.3.2.1 Disassembly and inspection.- At the conclusion of the Qualification tests, the units will be disassembled and inspected for excessive wear and defects.

4.3.3 Disposition of samples.- Upon completion of the tests, the units tested and the two untested units, together with test reports, shall be identified as required and forwarded to the activity responsible for qualification, designated in the letter of authorization from that activity (see 6.3). The test reports shall include a statement that the samples submitted have been subjected to and have complied with the Dielectric strength test (4.6.3). Each unit shall be accompanied by a set of spare brushes if brushes are used. Units shall be accompanied by the two copies of outline and detail assembly drawings thereof and by two printed copies of simple instructions with illustrations and diagram, if necessary, covering the installation of the units.

4.3.4 Procuring activity option - At the option of the procuring activity, any or all tests may be repeated.

4.4 Acceptance tests - The Acceptance tests shall consist of Individual tests and Sampling tests.

4.4.1 Individual tests - Each inverter shall be subjected to the following tests as described under "Test methods," except that temperature readings need not be taken for the Heating, Overload, and Commutation test specified in 4.6.2. Tests shall be performed in 3Ø Wye and 1Ø and may be performed in any sequence:

(a) Examination of product.

(b) Heating, overload and commutation (commutation applicable to brush type units only).

(c) Dielectric strength.

(d) Voltage and frequency regulation (room temperature only).

(e) Efficiency.

(f) Phase balance (3Ø units only).

(g) Output modulation.

In addition, the unit shall be subjected with the approval of the procuring activity, to any other tests specified herein which the Government Inspector considers necessary to determine unit compliance with the requirements of this specification.

4.4.2 Sampling tests.

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4.4.2.1 Overspeed - A quantity of 5 percent of the production of rotary inverters each month on each contract shall be selected at random and subjected to overspeed. (See 4.6.15).

4.4.2.2 Other sampling tests, if required, shall be as specified by the procuring activity. (See 6.2).

4.5 Test conditions - Unless otherwise specified, each test in this section shall be made under the following conditions.

4.5.1 Ambient temperature - Tests shall be performed at an ambient temperature of 25° ±15°C.

4.5.2 Altitude - Tests shall be run at approximately sea-level altitude. Test data shall include the dewpoint at which all altitude tests are conducted.

4.5.3 Warmup - Prior to the test, the unit shall be operated at maximum rated load for a sufficient time to reach a substantially constant temperature.

4.5.4 Voltage measurement - The input and output voltage shall be measured at the terminals of the unit.

4.5.5 Input voltage - Nominal input voltage as shown on applicable MS drawings shall be used.

4.5.6 3Ø and 1Ø Operation - Units shall be tested under rated loading conditions specified in 4.3.2 and 4.4.1.

4.5.7 Position - The units shall be mounted base down.

4.5.8 Loading -

4.5.8.1 Load power factor - Rated maximum power factor load shall be applied.

4.5.8.2 Multiple phase loading - A balanced load shall be used in multiphase tests.

4.5.8.3 Definition of load - Single phase load shall be considered the product of the RMS line current and the RMS voltage measured at the terminal of the machine. Three phase load shall be considered $\sqrt{3}$ times the product of the average RMS line current and the average RMS line voltage measured at the terminals of the machine.

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4.5.8.4 Location of load - The load for the unit shall be so located that it will not appreciably affect the ambient temperature.

4.5.9 Adjustment of frequency and voltage - No adjustment of frequency or voltage shall be made during any test.

4.5.10 Auxiliary equipment - All additional equipment indicated by the manufacturer as necessary to meet the requirements of this specification shall be used.

4.5.11 Test equipment - Test equipment, instruments, and meters used, shall be types generally accepted by industry for similar use.

4.6 Test methods -

4.6.1 Examination of product - Each unit shall be examined to determine compliance with all requirements of this specification for which there are no applicable tests.

4.6.2 Heating, Overload, and Commutation - Provision shall be made for determining input and output voltages, input and output currents, and other pertinent quantities. While the unit is cold, the resistance and temperature of the shunt fields, or other component whose temperature rise is indicative of the rise throughout the device, shall be determined for use in calculating the temperature rise (average) during continuous operation at rated load. The unit shall be considered to have reached its continuous operating conditions when the temperature rise of the shunt fields (or other component used to indicate temperature rise of the device) above the then existing ambient temperature is no more than 1°C during a period of 5 minutes. The ability of the unit to deliver and control rated power under the following continuous operating conditions shall be demonstrated.

4.6.2.1 Heating - The ability of the unit to deliver no less than minimum rated power, at maximum power factor, at rated input voltage for continuous operation shall be demonstrated. Immediately following the above run, the ability of the unit to start and to deliver no less than minimum rated power, at maximum power factor, for both minimum and maximum input voltage shall be demonstrated. The final temperature attained shall not be cause for rejection, provided the unit meets all other requirements of the specification.

4.6.2.2 Overload - The ability of the unit to deliver 200 percent rated load current for 5 seconds at the minimum voltage output specified on the applicable drawing shall be demonstrated at 26 volts input. The unit shall also demonstrate its ability to deliver no less than 250 percent rated load current under short circuit conditions.

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4.6.2.3 Commutation (Applicable to brush-type units only) - Immediately following the above heat runs, with the unit hot, the commutation of all brushes shall be observed over the input voltage range for rated load, half load, and no load. There shall be no more than fine pinpoint sparking during this test.

4.6.3 Dielectric strength - Dielectric strength tests may be conducted on the components before final assembly.

4.6.3.1 Circuits -

(a) Circuits which operate at voltages no greater than 50 volts shall be tested at 500 volts (RMS), 60 CPS for one minute or 600 volts (RMS), 60 CPS for one second.

(b) Circuits which operate at voltages in excess of 50 volts shall be tested at an rms voltage of 1250 volts at 60 CPS for one minute or 1500 volts for one second.

4.6.3.2 Capacitors - Capacitors shall be tested at a voltage equal to twice the voltage and at the frequency expected to be encountered in the intended application, for a period of one minute. (See 6.4.3.)

4.6.4 Voltage and frequency regulation - The regulated output voltage and frequency shall be measured under the conditions listed below. For each temperature condition, the voltage and frequency shall remain within the limits specified on the applicable drawing. However, in order to expedite testing units having a voltage adjustment, the voltage may be as much as 5 percent outside of the limits specified on the applicable detail specification or drawing, provided that the total deviation during test does not exceed the difference between the specified limits. The frequency for units having a frequency adjustment, may be outside of the limits by as much as 5 percent, provided the total deviation during test does not exceed the difference between the specified limits. There shall be no more than fine pinpoint sparking at the brushes during this test. When the applicable drawing places limits on the Freq/v ratio rather than on frequency and voltage separately, then the results of the above test will be interpreted accordingly and the specified limits must be met. Part 5, under (a), (b), and (c) below, may be omitted for inverters rated 10 va and less. At each different ambient temperature, the inverter shall be temperature stabilized prior to testing. The regulator shall show no signs of instability or harmful arcing.

(a) Voltage and frequency regulation at ambient temperature (25° ± 15°C):

(1) At minimum specified input voltage, the load shall be varied from no load to full load. The load shall be suddenly removed and applied for each observation. Sufficient data shall be taken to accurately plot the voltage and frequency regulation.

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(2) At rated input voltage, the preceding test shall be repeated.

(3) At maximum specified input voltage, the preceding test shall be repeated.

(4) At zero output current, the input voltage shall be gradually varied from minimum specified input voltage to maximum specified input voltage and returned. The output voltage shall not be adjusted, and data need be recorded only at minimum and maximum input voltage conditions.

(5) At 50 percent rated output current, the preceding test shall be repeated.

(6) At 100 percent rated output current, the preceding test shall be repeated.

(b) Voltage and frequency regulation at -55°C. Part (a) shall be repeated at -55°C.

(c) Voltage and frequency regulation at the maximum ambient temperature indicated by the applicable drawing. Repeat part (a) at the maximum temperature indicated by the applicable drawing.

During the course of the foregoing test, units having adjustable voltage or frequency controls shall demonstrate that the adjustment range and increments are as specified in 3.4.12.1 or 3.4.12.2, and that the phase rotation is as specified in 3.5.12.

4.6.5 Radio interference - At no load, half load, and full load, both conducted and radiated radio interference shall be measured using the test procedures and any applicable test instruments listed in Specification MIL-I-6181. Units shall not exceed the interference limits established by MIL-I-6181.

4.6.6 Efficiency - The power output and input of the unit shall be measured at full load, unity power factor, sea level, and nominal input voltage (28V). The efficiency shall be calculated under these conditions and shall be not less than that specified on the applicable MS drawing. Sufficient additional data shall be taken to determine over the load specified on the detail drawing.

4.6.7 Recovery time - A recording oscillograph shall be used to measure the output voltage transient when full rated load is applied and removed. It is desirable that the output voltage during load changes between full and no load recover to and remain within 10 percent of its steady-state condition within 0.1 seconds. It

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is required that the unit recover to and remain within 10 percent of its steady-state condition under such load changes within 0.3 second. The unit shall not develop sustained oscillations under any loading conditions.

4.6.8 Phase balance (3Ø units only). -

4.6.8.1 Definition - The percent unbalance of the phase voltage for a 3Ø unit shall be defined as 100 times the maximum deviation of the phase voltage from the average of 3Ø voltages, divided by the average of the 3Ø voltages.

4.6.8.2 Unbalanced loads - The effects of single Ø and unbalanced 3Ø loads on the balance of the 3Ø voltage output of the unit shall be determined as follows at both minimum and maximum input voltage. For units larger than 750 VA, the 2/3 unbalance condition shall not apply:

(a) When the unit is carrying no 3Ø load, a 1Ø, line-to-neutral unity-power-factor load requiring 1/3 and then 2/3 of the rated full-load output current (a load of approximately 1/9 and 2/9 of the full 3Ø rating of the unit) shall be connected successively to each phase.

(b) When the unit is carrying a unity-power-factor balanced 3Ø load requiring 1/3 of the rated full-load output current, a 1Ø, line-to-neutral unity-power-factor load requiring 1/3 and then 2/3 of the rated full-load output current (a load of approximately 1/9 and 2/9 of the full 3Ø rating of the unit) shall be connected successively to each phase.

(c) When the unit is carrying a unity-power-factor balanced 3Ø load requiring 2/3 of the rated full-load output current a 1Ø, line-to-neutral unity-power-factor load requiring 1/3 of the rated full-load output current (a load of approximately 1/9 of the full 3Ø rating of the unit) shall be connected successively to each phase. The maximum value of output voltage unbalance occurring during parts (a) through (c) above shall not exceed 4 percent.

4.6.8.3 Symmetry of construction - For all balanced loads irrespective of power factor, within the rating of the unit, the 3Ø output voltage shall not be unbalance by more than 3 percent.

4.6.9 Reduced input voltage -

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4.6.9.1 Starting contactor dropout voltage - The input voltage shall be reduced to 9 volts. The starting contactor, when incorporated, shall remain closed.

4.6.9.2 Reduced voltage operation - Units rated at 750 VA and below, with the exception of 10 VA units, shall demonstrate their ability to start and to operate continuously under full load at input voltages from 20 to 29 volts. Within this input voltage range, the output voltage shall remain between 100 and 120 volts and the frequency between 350 and 420 cps. Units rated above 750 VA shall start and operate with no load for at least 5 seconds at 22 volts with the output voltage remaining between 100 and 120 volts and the frequency between 370 and 410 cps.

4.6.10 Output modulation -

4.6.10.1 Output voltage modulation - The unit shall be operated from fully charged batteries or other sources approved by the activity responsible for qualification, and its total output voltage modulation determined by means of a calibrated oscilloscope or other means satisfactory to the procuring activity. The percent modulation shall not exceed 1 percent at no load, half load, or full load, and shall be computed according to the following formula:

$$\text{Percent modulation} = \frac{\text{Emax} - \text{Emin}}{\text{Emax} + \text{Emin}} \times 100$$

The unit shall not develop sustained voltage oscillations any of the above conditions of load. When an additional requirement covering the voltage modulation within a specified frequency band is included on the applicable drawing, the above formula shall apply over the specified band.

4.6.10.2 Output frequency modulation - The unit shall not develop sustained frequency oscillations under conditions of no load, half load, or full load.

4.6.11 Wave form - The crest factor and harmonic content of the output voltage per phase for 3 phase operation and across the 2 output lines for 1 phase operation shall be determined under the following conditions:

(a) At minimum input voltage specified on the applicable MS drawing for no load, half load, and full load (unity power-factor-steady-state resistive load).

(b) For steady-state-minimum-lagging power factor load specified on the applicable MS drawing, part (a) shall be repeated.

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(c) Tests (a) and (b) for rated input voltage shall be repeated.

(d) Tests (a) and (b) for maximum input voltage specified on the applicable MS drawing shall be repeated.

The crest factor obtained by the above test shall be 1.414 ± 10 percent, and the observed change in crest factor from no load to full load at any input voltage shall not exceed 10 percent. The value of any harmonic observed during the above test shall not exceed 5 per cent of the fundamental. This paragraph is applicable to inverters of 100 VA or larger.

The crest factor shall be determined by means of an oscillograph or peak and rms voltmeters. When voltmeters are used, the peak reading instrument shall be connected successively for each of the two possible polarities, and the crest factor shall be the ratio of the average of the two peak values of the reading of the rms voltmeters. The meters shall be of approved types. The harmonic content shall be determined by an approved analyzer.

4.6.12 Factors affecting voltage and frequency regulation -

4.6.12.1 Determination of total deviation - Prior to each of the tests (a) through (e) below, the voltage and frequency shall be at some value termed "standard" for the purpose of this test, which lies between the limits specified on the applicable drawing and which has been determined after voltage, frequency, and temperature stabilize. The maximum deviation of voltage and frequency in both positive and negative directions form the "standard" shall be recorded for each test. The following tests shall be conducted at full rated load unless otherwise specified:

(a) *Ambient temperature:* The ambient temperature shall be decreased from 25° ± 15° to -55°C, increased to the maximum ambient temperature(s) indicated by the applicable drawing, and then checked at 25° ± 15°C. Each temperature shall be held for at least 1 hour or until the regulated voltage and frequency stabilize as indicated by readings taken at 10-minute intervals. The ability of the unit to continuously deliver rated output at ambients between -55°C and the maximum ambient temperature(s) indicated by the applicable drawing shall be demonstrated, and the value of the voltage and frequency recorded. Tests may be made at any ambient temperature between the limits specified, at the option of the activity responsible for qualification.

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(b) Power factors: The load power factor shall be decreased from maximum to both the minimum lagging and the minimum leading value specified on the applicable drawing. Tests will be made at any other load power factors between minimum lagging and minimum leading at the option of the activity responsible for qualification. Each load power factor shall be held until regulated voltage and frequency stabilize.

(c) Altitude: The pressure altitude shall be varied from approximately sea level to conditions approximating the maximum altitude given on the applicable detail specification or drawing. Tests shall be made at intermediate altitudes at the option of the activity responsible for qualification. Each altitude shall be held until the regulated voltage and frequency stabilize. The ambient temperature at each altitude shall be within 5° C of the maximum on Standard MS33543 for the range indicated by the applicable detail specification or drawing.

(d) Position: The unit shall be rotated by increments of not more than 90 degrees from the normal horizontal position, 360 degrees about each major axis, except the vertical axis.

(e) Warmup: Only the maximum voltage and frequency deviation value obtained from the tests below shall be used in computing total deviation:

(1) The unit shall be soaked at an ambient temperature of -55°C for a period of at least 8 hours during which time no power shall be connected to the unit. At the end of this period, the input terminal shall be energized and the unit shall be started under full rated load. Readings of regulated voltage shall be recorded at intervals of no longer than 2 minutes until the voltage and frequency stabilize. The voltage shall not exceed the maximum values shown on figure 2. Readings taken during the first 2 seconds shall not be used in the computation of the total deviation.

(2) The above test (1) shall be repeated with no load on the unit.

(3) The above test (1) shall be repeated at the maximum ambient temperature(s) indicated by the applicable drawing.

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(4) The above test (2) shall be repeated at the maximum ambient temperature(s) indicated by the applicable drawing.

For both voltage and frequency, the five values of maximum positive deviations recorded from tests (a) through (e) above shall be added together and the five values of maximum negative deviations shall be added together. To these sums add respectively the maximum positive and the maximum negative deviations owing to changes in load and input voltage at room ambient which were recorded during the test specified in 4.6.4(a) (1), (2), or (3). The total positive and the total negative deviations thus obtained shall be added algebraically to the standard values of voltage and frequency separately, and the unit shall be considered acceptable if the voltage and frequency values thus obtained remain within the limits specified on the applicable drawing, except that to expedite testing, the same 5 percent allowance may be made for voltage and frequency as described in 4.6.4. When the applicable drawing places limits on the freq/v ratio, the results of the above tests will be interpreted accordingly and the specified limits must be met.

4.6.12.2 Regulation at worst indicated conditions - If the unit is found to be not acceptable under the total deviation test specified in 4.6.12.1, further tests shall be run to determine regulation at the combination of the worst indicated conditions as follows: To expedite testing, the same 5 percent allowance may be made for voltage and frequency as described in 4.6.4.

(a) The worst indicated conditions of temperature, altitude, operating position, input voltage, load power factor, and magnitude of load to be used for this test shall be those conditions in the tests specified in 4.6.12.1 and 4.6.4(a) (1), (2), or (3) where the inverter was found to have the maximum positive voltage deviations. The unit shall be placed in the worst indicated operating position and soaked for at least 24 hours (not operating) at worst indicated ambient temperature, after which the pressure altitude shall be adjusted to its worst indicated value. During this test, the temperature-altitude conditions shall not exceed the limits of Standard MS33543. Following this, the unit shall be started by connecting power to the input terminals, with the "on-off" switch (if provided) in "on" position. The worst indicated load (no load, or load at worst indicated power factor) shall be immediately applied. Readings of voltage and frequency shall be recorded immediately after starting and every 2 minutes until they stabilize. The reading taken immediately after starting will be permitted to exceed the upper limit shown on the applicable drawing or detail specification by as much as 5 percent, but all other readings shall be within the specified limits.

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(b) The above test (a) shall be repeated for conditions found to have maximum negative voltage deviations. The output voltage and frequency shall be adjusted (if adjustment means are provided) to the same value used in test (a) above, prior to starting this test.

(c) The above test (a) shall be repeated for conditions found to have maximum positive frequency deviations. The output voltage and frequency shall be adjusted (if adjustment means are provided) to the same value used in test (a) above, prior to starting this test.

(d) The above test (a) shall be repeated for conditions found to have maximum negative frequency deviations. The output voltage and frequency shall be adjusted (if adjustment means are provided) to the same value used in test (a) above, prior to starting this test.

If voltage and frequency remain within the specified limits during tests (a), (b), (c), and (d) above, the unit shall be considered to have satisfactorily fulfilled the requirements of the test for factors affecting voltage regulation, even though the voltage values computed by using total deviations specified in 4.6.12.1 did not remain within the specified limits.

4.6.13 Operating position - During the course of other tests, it shall be ascertained that the operation of the unit in any position does not adversely affect power input, heating, lubrication, and other operating characteristics for all units.

4.6.14 Acceleration - The acceleration test shall be performed in accordance with procedure 3 of Specification MIL-E-5272. Deviation in controlled voltage and frequency, at both full load and no load, produced by acceleration in any direction shall not exceed 3 percent. The regulator controls shall show no sings of instability.

4.6.15 Overspeed - (Applicable only to rotary units.) The unit shall demonstrate its ability to operate at 25 percent above the rated speed at no load for 5 minutes without mechanical failure. Immediately following this test, the unit shall pass the test specified in 4.6.4(a) (2).

4.6.16 Starting - By the addition of external resistance in the line or other suitable means the input voltage shall be made to drop approximately 4 percent for each multiple of full load input current drawn. For example: If the full load input current as computed from minimum allowable unit efficiency is found to be 8 amperes, and 20 amperes are drawn at starting, then the voltage supplied to the unit shall drop to $100 - \frac{4 \times 20}{8}$ or 90 percent nominal input voltage.

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4.6.16.1 Starting current - The starting current shall be measured using an oscillograph, while starting the unit at ambient room temperature. The current shall not exceed 1500 amperes or 1000 percent of full load current (whichever is the least) as computed from minimum allowable unit efficiency specified on applicable MS drawing. The current shall return to and stay within 300 percent of full load within 0.3 second after first application of input voltage. The starting current oscillogram shall indicate a definite current peak and shall show no tendency for current to remain at peak value.

4.6.16.2 Starting voltage and frequency - The oscillograph record taken in 4.6.16.1 shall show the output frequency and voltage. During the starting cycle, output shall not exceed the values shown on figure 2. Twenty-five seconds after starting, the voltage and frequency shall reach and remain within the limits specified in the detail specification or drawing.

4.6.16.3 Cyclic starting conditions - The unit shall be subjected to 50 consecutive 5-minute cycles of 1 minute operation, and 4 minutes inoperative. The voltage and frequency shall be within the limits specified in the applicable MS drawing within 1 second after each energizing operation, except the first. Upon completion of this test, the unit shall pass the test specified in 4.6.4(a) (2). No parts shall show signs of excessive burning or wear.

4.6.17 Endurance - The unit shall be operated for 200 hours, during which time full-rated load shall be switched alternately on and off for 5 second periods. During this test, readings shall be taken at no more than 12 hour intervals to demonstrate that the output voltage and frequency do not drift more than 2 percent during the entire test or any observed interval thereof. Following this test, the unit shall pass the tests specified in 4.6.4(a) (2) and 4.6.7. No part of the unit shall show excessive wear.

4.6.18 Shock - The unit shall be subjected to the Shock tests of Specification MIL-E-5272, Procedure V, Equipment Operation. There shall be no mechanical damage as a result of the foregoing shock tests. Following these tests, the unit shall pass the test specified in 4.6.4(a) (2) of this specification. The unit shall then be subjected to the Shock tests of Specification MIL-E-5272, Procedure V, Equipment Crash-Safety.

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4.6.19 Vibration -

4.6.19.1 Inverters weighing less than 35 lbs - The inverter shall be subjected to durability and performance tests as follows:

(a) Durability - The inverter shall be operated at full load during this test. The inverter shall be subjected to harmonic or circular motion applied to the mounting base. The amplitude shall be .015 inch (maximum total excursion of 0.03 inch). The frequency of the vibration shall be established by cyclic operation from approximately 10 cps to 55 cps and back to 10 cps. The duration of one such cyclic operation shall be between 1 and 5 minutes. The inverter shall be vibrated continuously for 2 hours in the direction of each of the 3 major axes. At the end of the test period, the inverter shall be thoroughly inspected for damage or other defects resulting from the vibration test. Following this test, the inverter shall pass the test specified in 4.6.4(a).

(b) Performance - The inverter shall be subjected to simple harmonic or circular motion applied to the mounting base. The amplitude shall be 0.015 inch (maximum total excursion of 0.03 inch), up to 60 cps; above 60 cps the acceleration shall be limited to 10g. The inverter shall be vibrated over the frequency range of 20 to 500 cycles per second along each of its major axes. The vibration excursion shall take 12 to 15 minutes with no one frequency held longer then 3 minutes. The output voltage modulation at any vibrating frequency measured in the manner previously specified shall not exceed 2 percent. Following this test, the inverter shall pass the test specified in paragraph 4.6.4(a).

4.6.19.2 Inverters weighing more than 35 lbs - The inverter shall be subjected to the following tests. The maximum acceleration over the entire vibration envelope shall not exceed 5 g's during the following tests.

(a) Resonance - The inverter shall be operated at full load during this test. Resonant frequencies of the inverter shall be determined by varying the frequency of applied vibration slowly through the specified range of frequencies at vibratory accelerations not exceeding those shown in Figure 3. Individual

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resonant frequency surveys shall be conducted with vibration applied along each of any set of three mutually perpendicular axes of the inverter. The inverter shall be vibrated at the indicated resonant conditions for the periods shown in the vibration test schedule (Table I) and with the applied double amplitude or vibratory acceleration specified in Figure 3. These periods of vibration shall be accomplished with vibration applied along each of three mutaully perpendicular axes of vibration. When more than one resonant frequency is encountered with vibration applied along any one axis, the test period may be accomplished at the most severe resonance or the period may be divided among the resonant frequencies, whichever is considered most likely to produce failure. When resonant frequencies are not apparent within the specified frequency range, the speciman shall be vibrated for periods twice as long as those shown for resonance in the vibration test schedule at a frequency of 55 cps and an applied double amplitude of 0.060 inch. Following this test, the unit shall pass the test specified in 4.6.4(a).

(b) Cycling - A vibration test shall be conducted with the frequency cycling between 10 and 500 cps in 15 minute cycles at an applied double amplitude of 0.036 inch or an applied acceleration of ± 5g whichever is the limiting value. Vibration shall be applied along each of three mutually perpendicular axes according to the vibration test schedule (Table I). Following this test, the unit shall pass the test specified in 4.6.4(a).

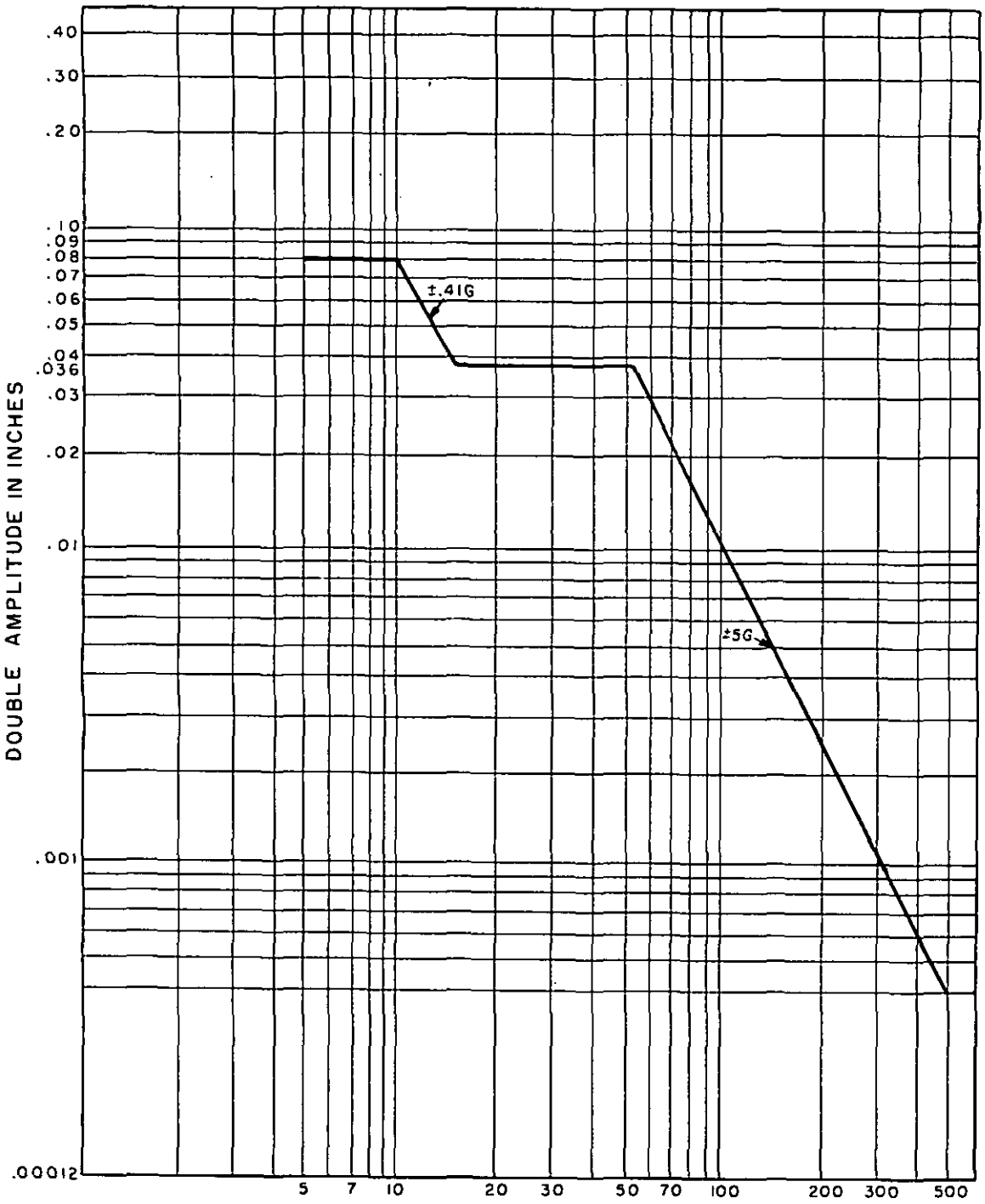
TABLE I

VIBRATION TEST SCHEDULE

(Times shown refer to one axis of vibration)

Type	Vibration at Room Temp	Vibration at 160°F (71°C)	Vibration at -65°F (-54°C)
Resonance	60 minutes	15 minutes	15 minutes
Cycling	60 minutes	15 minutes	15 minutes

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RANGE CURVE FOR VIBRATION TESTS

FIGURE 3

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4.6.20 Performance - The unit shall be operated under the following conditions at full load unless otherwise indicated. New brushes, when used, may be installed for this test. The unit shall pass the voltage and frequency regulation test specified in 4.6.4(a) after the completion of each cycle. This test shall consist of 4 continuous cycles conforming to the following schedule with the maximum load, temperature, and altitude as specified by the applicable detail specification or drawing:

6 hrs at sea level

6 hrs at 20,000 ft.

6 hrs at 35,000 ft.

6 hrs at 50,000 ft.

6 hrs at maximum altitude

The unit shall be inspected for signs of overheating, excessive deterioration, and wear. The total wear for all components, except brushes, shall not exceed 12 percent of the allowable wear for the 4 cycles. The brush wear shall not exceed 24 percent of the allowable wear for the 4 cycles. There shall be no evidence of grooving or other surface damage to the face of the brush. Failure during this test shall be cause for rejection.

4.6.21 Life - The unit shall be operated for 1000 hours. For the first 25 hours, it shall deliver rated current while operating at the maximum ambient temperature(s) indicated by the applicable MS drawing. For the remainder of the test, the unit shall operate in an ambient temperature of 25° ± 15°C. Units having an output of less than 100 va shall operate with the mounting base in a position other than directly below the unit during 500 hours of this test. Data shall be recorded and the unit shall be brought to a complete stop and started under full load within 3 minutes at least every 25 hours during the test to demonstrate that the output voltage and frequency do not drift more than 2 percent during any 100-hour period and no more than 4 percent during the entire test or any observed interval thereof. During the last hour of the first 25-hour period and following this test, the unit shall pass the test specified in 4.6.4(a). For this Regulation test, the voltage and frequency may be adjusted if adjusting means are provided externally to compensate for the Life test. No lubrication or maintenance of any sort, other than that permitted by 3.4.12.1, 3.4.12.2, and 3.5.5 shall be allowed during this test.

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4.6.22 Humidity - The unit shall be subjected to the Humidity tests, Procedure I, of Specification MIL-E-5272. Immediately following this test, the unit shall pass the test specified in 4.6.4(a) (2).

4.6.23 Salt spray - The unit shall be subjected to 25 hours of salt spray in accordance with Federal Test Standard 151, Method 811. Following this test, the unit shall be washed and dried for 15 to 20 hours and then shall meet the requirements of 4.6.4(a) (2). There shall be no evidence of excessive corrosion.

4.6.23.1 Synthetic Sea Water Solution - The synthetic sea water solution shall be formulated as prescribed as follows:

Salt	Grams Per Liter
Sodium chloride	24.540
Magnesium chloride	11.110
Sodium sulphate	4.094
Calcium chloride	1.159
Potassium chloride	0.695
Sodium bicarbonate	0.201
Potassium bromide	0.101
Boric acid	0.027
Strontium chloride	0.042
Sodium fluoride	0.003

Adjust the pH of this solution, if necessary, to 8.2 with 0.1 normal sodium carbonate solution.

4.6.24 Sand and dust - The unit shall be subjected to the Sand and Dust tests, Procedure I, of Specification MIL-E-5272, except that the test shall be run at sand and dust velocities of both 200 ± 100 FPM and 2300 ± 500 FPM. The 8-hour test cycle shall be composed of 30-minute operative (no load) and 90-minute inoperative periods. The total operative test duration shall be 2 hours and the inoperative test duration shall be 6 hours. At the end of the test, the unit shall be removed from the test chamber and subjected to the test specified in 4.6.4(a) (2).

4.6.25 Fungus - The unit shall be subjected to the Fungus resistance tests, Procedure I, of Specification MIL-E-5272. Immediately following this test, the unit shall meet the test requirements of 4.6.4(a) (2).

4.6.26 Explosion-proof - When specified by the applicable detail specification or drawing, an Explosion-proof test conforming to the indicated procedure in Specification MIL-E-5272, shall be performed.

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4.7 Materials test specimens - Test specimens shall be furnished when requested by the Government for check tests at a Government laboratory to determine conformance with the applicable specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - The inverters shall be unit packaged in accordance with Specification MIL-P-7936, level A or C as specified in contract or order.

5.1.1 Level A - For level A packaging, preservation shall be in accordance with Specification MIL-P-116, Method II, without preservation compounds, using metal reusable containers otherwise specified.

5.1.2 Level C - For level C the inverter shall be preserved and packaged individually in accordance with manufacturer's commercial practice.

5.2 Packing - Packing shall be in accordance with Specification MIL-P-7936, level A, B, or C as specified in contract or order.

5.3 Marking - Interior and exterior containers shall be marked in accordance with Specification MIL-P-7936 and Military Standard MIL-STD-129.

5.3.1 Reinspection dates - Reinspection date marking shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use - This specification covers the requirements for inverters, which are used as devices to convert direct current electrical energy into alternating current electrical energy. They are intended for continuous duty on aircraft.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Class

(c) MS part number

(d) Sampling tests when required

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(e) Required level of packaging and packing

(f) Type of container

(g) Reinspection date marking and any special marking

(h) Installation instructions, handbooks, or manuals when required

6.3 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Chief, Bureau of Naval Weapons, Department of the Navy, Washington 25, D. C. and information pertaining to qualification of products may be obtained from that activity.

6.3.1 Reproducible copies - Upon notification of qualification approval, the manufacturer shall provide two reproducible copies of outline and detail assembly drawings. Any changes in the reproducible copies from those submitted with the qualification samples shall be indicated in detail.

6.4 Miscellaneous Notes -

6.4.1 Phase voltage - Phase voltage shall be considered to be line-to-neutral.

6.4.2 Alternating voltage - Alternating voltages used in this specification shall be root mean square (RMS) values.

6.4.3 Dielectric strength test for capacitors - The dielectric strength test for capacitors described in Section 4 is not a test of ultimate dielectric strength and is not presented as a design criterion. The purpose of this test is to assure the satisfactory performance of individual capacitors under operating conditions. Capacitors used shall be suitable for the purpose, and shall conform to applicable Government specifications. Successful performance of the dielectric strength test does not relieve the contractor of the responsibility of providing adequate safety factors consistent with accepted design practices.

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Custodians:

Army - MO
Navy - WP
Air Force - 11

Preparing Activity:

Navy - WP

(Project No. 6125-0088)

Review Activities:

Army - MO, EL
Air Force - 11

User Activities:

Army - MI

Review/User information is current as of the date of this document. For further coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-I-7032, Inverter, Aircraft; proposed Revision F

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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